

Scalability & Future Plan

Date	07 July 2026
Team ID	SWTID-2026-8075
Project Name	Voice Based Concept Understanding Analyser(VBCUA)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to uploaded audio files	Reduces real time usability	High
2	Supports a limited set of reference concepts	Restricts evaluation scope	Medium
3	Single-user/local deployment	Not suitable for many concurrent users	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local Streamlit application	Deploy on cloud with load and multiple instances
2	Data Storage	Local storage for uploads/reports	Use cloud database and object storage
3	Performance	Sequential processing	Optimize models, caching, and asynchronous processing
4	Security	Basic local access	Authentication, HTTPS, and encrypted storage

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Real time microphone recording	1-2 months	Improved user experience
Phase 2	Multi-language speech recognition	2-3 months	Wider accessibility
Phase 3	User authentication & speech recognition	3-4 months	Personalized tracking
Phase 4	Cloud deployment & advanced AI feedback	4-6 months	Scalable intelligent platform